Source and Diffuse-LyC Photon-Energy Sampling: Code Comparison

1. Scope

This report is based on direct source inspection of MOCASSIN 2.02.73.2, `Wood_Codes/ionize`, CMacIonize-master, `TORUS/torus/src`, and MoCHII. For equal-energy packets the source photon CDF is normally proportional to $L_\nu/(h\nu)$. A free-bound LyC CDF also follows the Milne relation and includes the inverse photoionization cross section.

2. Comparison

Code	Source energy	Diffuse LyC energy	Result
MoCHII	Continuous input-spectrum inverse CDF; sampled energy retained in continuous mode.	Temperature-indexed Milne CDF for each ground continuum, inverted once after interpolating in $\log T$; separate He I cascade branches.	Exact transport energy; exact ground-continuum shape. He I cascade exact in branch weights, two-photon shape and 584 Å escape-probability branching, whose products are then transported (§7.).
MOCASSIN	Inverse CDF on global frequency grid.	Cell-emissivity CDF on the same grid.	Flexible tabulated emissivity, but grid-valued packets.
Wood ionize	Log-interpolated Planck/atmosphere CDF.	Temperature-indexed, cross-section-weighted Milne CDF.	Explicit H/He free-bound continuum shape.
CMacIonize	Numerical spectrum-specific CDF.	Dedicated H and He LyC spectra selected by reemission branches.	Modern event-by-event physical reemission.
TORUS	Tabulated wavelength CDF, linear in selected interval.	No active HII diffuse-LyC reemission path found.	Generic continuum sampler only.

3. MoCHII

`src/energy_sampler_mod.f90` constructs the source CDF and `src/ionizing_field_mod.f90` launches packets. `gen_ion_photon` and `gen_ion_photon_qmc` (`src/ion_band_mod.f90`) store the sampled energy in `photon%energy_eV` (continuous energy has been the only transport mode since 2026-08-01, when the `ion_energy_mode` switch and `assign_ion_packet_energy` this report first named were removed); `ion_packet_mod.f90` and `ion_score_mod.f90` evaluate H, He, and metal physics at that energy. The ionizing-band index is diagnostic only. `src/diffuse_mod.f90` builds a leaf–channel CDF each iteration. Its H II→H I(1s), He II→He I(1^1S), and He III→He II(1s) channels use $\alpha_1 = \alpha_A - \alpha_B$ and, since 2026-07-29, sample the Milne photon spectrum $p(E) \propto E^2\sigma_{\rm pi}(E)e^{-(E-E_{th})/kT_e}$ through `src/milne_`

`recomb_spectrum_mod.f90`, selected by `par%diffuse_energy_model = 'milne'` (the default). The tables are built once in $x = (E - E_{th})/kT$, interpolated in $\log T$, and inverted once; both pseudorandom and Sobol launches call the same routine, and the Sobol dimension assignment is unchanged. Because packets are equal-energy and their count comes from the channel's energy luminosity, they are drawn from $E\,n(E)$ rather than $n(E)$, which makes the represented photon rate exactly $n_e n_{\rm ion} \alpha_1 V$; the luminosity still uses the number-weighted mean $\langle E \rangle_n$. Start-up reports α_1 recovered from the same spectral integral against $\alpha_A - \alpha_B$ without rescaling either; since the α_1 correction of §11. that report reads 0.01% instead of 3.44%. `exponential` ($E = E_{th} + kT_e[-\ln u]$) is retained for reproducing earlier runs and `threshold` as a sensitivity bracket; neither is a physical replacement. The optional He I cascade now weights its 19.82-eV, 21.22-eV and two-photon branches by effective recombination coefficients instead of the AGN3 statistical weights, and draws its ionizing two-photon energies from the Drake et al. (1969) shape rather than uniformly. At 8000 K the branch fractions are 0.762 : 0.076 : 0.162 against the statistical 0.750 : 0.083 : 0.167, a temperature-dependent correction rather than a reordering, and the channel radiates 19.223 eV in 0.9663 photons per case-B He II recombination instead of the 19.200 eV in 0.9634 photons of the statistical weights with a flat two-photon spectrum (+0.12% in energy, +0.30% in photon number), before the 584 Å branching of §7. is applied on top. Section 7. compares this cascade, and the fate of its 584 Å photon, with the three reference codes. (A fifth channel, the He III case-B cascade — free-bound into $n \geq 2$, He II Lyα and the 2^2S two-photon pair, `src/heii_cascade_mod.f90`, `par%heii_diffuse` — was added on 2026-08-09, after this report; it is on by default with the transported continuum when `par%eion_max` ≥ 40.813 eV.)

4. MOCASSIN

In `source/photon_mod.f90`, `newPhotonPacket` calls `getNu2(inSpectrumProbDen,...)` for stellar photons. `getNu` and `getNu2` invert cumulative arrays and return an integer frequency pointer. `continuum_mod.f90` builds the continuum representation and `setLdiffuse` accumulates diffuse luminosity. Diffuse continuum packets use the same pointer/CDF mechanism. Thus arbitrary free-bound emissivities can be tabulated, but packet frequency is a global-grid value rather than a continuous in-bin variate.

5. Wood ionize

`planckset.f` prepares the source CDF; `nuset.f` locates a uniform deviate in `bbcdf` and logarithmically interpolates `lognu`. `probinit.f` explicitly constructs photon CDFs, $\int j_\nu/\nu\, d\nu$. At every `Tarray` point it calls `phfit2` and uses $j_{HI} \propto \nu^3 \sigma_{HI}(\nu) \exp[-157919.667(\nu - 1)/T]$, with an analogous He I expression above 1.81 Ryd. It integrates j_ν/ν to `pH` and `pHe`. `nuemit.f` inverts the temperature-indexed H/He CDFs and the He two-photon CDF; `mcionize.f` selects the reemission branch using `probset.f`. This is a Milne-grid sampler, though its output is a grid frequency.

6. CMacIonize

`PlanckPhotonSourceSpectrum.cpp` tabulates $\nu^2/(e^{h\nu/kT} - 1)$ over 1–4 Ryd, integrates a CDF, and interpolates in logarithmic CDF–frequency space in `get_random_frequency`. `PhysicalDiffuseReemissionHandler.cpp`

chooses absorption with $n_i\sigma_i$ weights and applies temperature-dependent reemission probabilities. It invokes `HydrogenLymanContinuumSpectrum` and `HeliumLymanContinuumSpectrum`, and separately treats the 19.8-eV and 56% ionizing He two-photon branches. These dedicated spectra receive cross-section objects, implementing the same physical design as Wood ionize.

7. He I cascade and the 584 Å photon

The reemission branch that `probset.f` sets up above is where the four codes differ most, and it holds two separate decisions: which $n = 2$ term a case-B He II recombination passes through, and what becomes of the 2^1P photon, He I Lyα at 584 Å = 21.22 eV.

Branch coefficients. `probset.f` sets four effective recombination coefficients from Benjamin, Skillman & Smits (1999) Table 1 with J. S. Mathis — $1.54 \times 10^{-13} T_4^{-0.486}$ to 1^1S, $2.10 \times 10^{-13} T_4^{-0.381}$ to 2^3S, $2.06 \times 10^{-14} T_4^{-0.451}$ to 2^1S and $4.17 \times 10^{-14} T_4^{-0.695}$ to 2^1P — and normalizes by their sum, for the reason the source states: "different T-dependences in fitting formulae". `PhysicalDiffuseReemissionHandler` in CMacIonize is a line-for-line reimplementation of that choice, with the same four coefficients, the same sum normalization and the same cumulative channel order, and cites Wood, Mathis & Ercolano (2004) §3.3. MOCASSIN does not resolve the split at all: its nebular continuum uses $6.23 \times 10^{-14} T_4^{-0.827}$ for the He I two-photon emissivity under the stated assumption that "all HeI singlets finally end up in the 2^1S". That coefficient is the singlet total, $2.06 \times 10^{-14} + 4.17 \times 10^{-14} = 6.23 \times 10^{-14}$ exactly, refitted as a single power law, so it is the full-conversion limit written as a constant.

The 584 Å photon. `mcionize.f` does not propagate it. It draws $p_{\rm Hots} = [1 + 0.77(100/\sqrt{T})\,x({\rm He}^0)/x({\rm H}^0)]^{-1}$ and with that probability absorbs the photon on the spot into the hydrogen Lyman continuum, depositing $21.22 - 13.60 = 7.62$ eV; otherwise the 2^1P converts to 2^1S (its 2.06 μm decay) and the atom decays as the two-photon continuum, from which an ionizing photon emerges with probability 0.56. The same $p_{\rm Hots}$ enters the ionization balance (`h0find.f`) and the heating (`ioneng.f`). CMacIonize writes $p_{\rm Hots} = \sqrt{T}\,x({\rm H}^0)/[\sqrt{T}\,x({\rm H}^0) + 77\,x({\rm He}^0)]$, which is algebraically the same expression, and also destroys the photon at the emission site.

MoCHII. Since 2026-07-30 the branch fractions come from the same three $n = 2$ coefficients under the same sum normalization, so all three are drawn from one source: a ratio of coefficients with unrelated temperature dependences is not a branch fraction, and closing one branch against an external α_B would place the mismatch between two data sets in whichever branch is solved last. The sum is $0.91\,\alpha_B$(He II) at 8000 K and 0.985 at 10^4 K on the current rates (0.90 and 0.97 when this report was written, before the Milne α_1 correction of §11. lowered α_B(He I) by 1.9%), and that shortfall is reported at start-up instead of being forced into a branch. The two-photon energies follow the Drake table (0.55644 ionizing photons and 8.9646 eV per decay, in-band mean 16.110 eV) rather than the flat surrogate (0.56, 17.109 eV). The 2^1P branch is split by the same escape probability as in the two reference codes, written in this module as $p_{584} = [1 + 77\,(y_{\rm He}/0.1)\,x({\rm He}^0)/(\sqrt{T}\,x({\rm H}^0))]^{-1}$ (`hei_584_ionizes_H`; the factor $y_{\rm He}/0.1$ = `par%abund_He`$/0.1$ since 2026-09-26, because WME04's 0.77 assumes He/H = 0.1, and it is 1 at that abundance); the gate `tests/hei_cascade/check_hei_584.f90` shows this form agreeing with Wood's and CMacIonize's algebraically distinct ones to 2.2×10^{-16}, both limits exact, and $p_{584} = 1$ reproducing the previous branch energies. Because p_{584} depends only on the emitting leaf's own state it enters the branch weights of the emission table, so no

Code	$n = 2$ branch coefficients	2^1P (584 Å) fate	Transported?
Wood ionize	BSS99 + JSM, four coefficients, normalized by their sum	$p_{\rm Hots}$ ionizes H on the spot, $1 - p_{\rm Hots}$ converts to 2^1S two-photon (ionizing fraction 0.56)	No (on the spot)
CMacIonize	Identical set and sum normalization, citing WME04 §3.3	Identical branching, written in an algebraically equivalent form	No (on the spot)
MOCASSIN	Singlet total $6.23 \times 10^{-14} T_4^{-0.827}$ only	All singlets assumed converted to 2^1S	Emissivity only
MoCHII	Same three $n = 2$ coefficients, normalized by their sum; $\sum \alpha/\alpha_B$ reported, not forced	Same escape-probability split, from the leaf's own state; converted decays radiate 20.62 eV with the Drake two-photon shape, and `nebcont_mod` uses the same split	Yes (AMR grid)

Sobol dimension was added and the assignment is unchanged. A converted decay radiates 20.62 eV as the two-photon pair rather than 21.22 eV, the missing 0.60 eV leaving as the non-ionizing 2.06 micron $2^1\mathrm{P} \rightarrow 2^1\mathrm{S}$ line. The same split sets the He I two-photon rate of `nebcont_mod`: since 2026-09-26 it takes the 2^1S decays from `hei_twophoton_decays` (`hei_recombination_photons_mod`), $n_e n(\mathrm{He}^+)\,\alpha_B(\mathrm{He\,II})\,[f_{1S} + (1 - p_{584}) f_{1P}]$ from the same branch fractions, plus the collisional transfers out of 2^3S when `par%hei_metastable_coupling` is on (the default); in the $p_{584} \rightarrow 0$ limit its singlet part is the branch-fraction form of MOCASSIN's singlet total. (From 2026-07-30 to 2026-09-26 it was $\alpha(2^1\mathrm{S}) + [1 - p_{584}]\alpha(2^1\mathrm{P})$ with the BSS99 fits in absolute units, which reduces to MOCASSIN's 6.23×10^{-14} exactly.) the transported packets and the written continuum therefore describe the same nebula, closing an internal disagreement in which the two modules had sat at opposite limits of the same decay. (An earlier version of this report read 6.23×10^{-14} as the 2^1S coefficient alone and closed 2^1P against α_B, obtaining 0.746 : 0.224 : 0.031 and an argument that case-B trapping inverts the singlet split. That reading is withdrawn: the coefficient is the singlet total.)

Where MoCHII stands. Of the four codes only MoCHII both applies the branching and then transports what comes out of it: Wood and CMacIonize destroy the 584 Å photon where it is emitted, so its deposition is local by construction, and MOCASSIN carries the decay in an emissivity only. MoCHII draws the same split from the same closure and hands the surviving 584 Å packets and the two-photon photons to the grid, so where their energy is deposited is resolved rather than assumed. The branch content is no longer the weak part of that arrangement: the branch fractions, the two-photon shape and the 584 Å fate all come from the same sources the reference codes use. Evaluated on the converged production grids and weighted by the 2^1P emission rate, $p_{\rm Hots}$ is 0.117 at HII40 and 0.0044 at HII20, so 88% and 99.6% of the branch converts to the two-photon continuum, and the channel loses 7–8% of its photons and softens. What remains here is explicit 584 Å resonance-line transfer, for which $p_{\rm Hots}$ is an escape-probability closure valid for an optically thick, static, dust-free line, and collisional redistribution among the $n = 2$ terms at high density.

8. The He I recombination coefficient, code by code

The cascade of §7. distributes a rate that each code fixes separately, and that rate is where the four codes disagree by far the most. Reading each code's own data rather than its published description gives, for α_A(He I) at 10^4 K:

Code	How α_A(He I) is obtained	10^{-13} cm^3 s^{-1}	vs. Cloudy
Cloudy c25.00	Level-by-level Milne from its own $\sigma_{\rm pi}$, total cached	4.3583	—
MoCHII	Badnell RR total + DR	4.3719	+0.3%
Wood `ionize` / CMacIonize	Sum of the four BSS99 *effective* coefficients	4.2630	−2.2%
MOCASSIN	Verner & Ferland 1996 total fit	**4.8394**	**+11.0**%

Three treatments of the cascade, none of them truncated. The spread is not a disagreement about how far the recombination sum runs. Cloudy sums explicitly over resolved levels; MoCHII and MOCASSIN use totals over all n (Badnell, VF96); Wood and CMacIonize use *effective* coefficients, in which the cascades from higher levels are already folded into the $n = 2$ terms — which is exactly why §7. must normalize those four coefficients by their sum rather than treat them as absolute rates. All three are complete, and MoCHII's Badnell total agrees with Cloudy's explicit level sum to 0.3%, which a truncation on either side would break. The 20-level list in Cloudy's `source/sanity_check.cpp` is a unit test of threshold cross sections, not the extent of its recombination sum.

The MOCASSIN value is decoded from `data/radrec.dat`, replicating the read order of `update_mod.f90` lines 1707–1735. Two controls validate that decode on the channels whose answer is known independently: H I 4.1923 against Cloudy's 4.1642×10^{-13} (+0.7%) and He II 2.1880 against 2.1817×10^{-12} (+0.3%).

The electron count is the control that settles it. The one-electron species agree to 0.7% (H I) and 0.3% (He II), because a hydrogenic system has an exact analytic solution and there is nothing for two calculations to disagree about. The two-electron He I spreads by 11–13%. The disagreement therefore lives in the atomic data for a two-electron atom — screening and the singlet/triplet structure — and not in the extent of anybody's cascade sum.

Three of the four cluster. Within He I the values split three-to-one: Cloudy's level sum 4.3583, Badnell 4.3719 and BSS99 4.2630×10^{-13} all lie within 2.5% of each other, and Verner & Ferland (1996) alone sits at 4.8394×10^{-13}. MOCASSIN applies its `radRecFit` uniformly with no special case for H or He (`source/update_mod.f90:1636--1639`, whose own comment names "H-like, He-like, Li-like, Na-like — Verner & Ferland, 1996, ApJS, 103, 467"), and adds dielectronic recombination only for $Z_{\rm elem} \geq 3$, so its He I rate is the VF96 radiative value alone. Offered as reasoning rather than as a verified claim: VF96's He-like fits lean on hydrogenic scaling for the excited levels, and $Z = 2$ is where that approximation is worst, since neutral helium is exactly where screening and the singlet/triplet splitting matter most.

A caution for anyone validating against MOCASSIN. MOCASSIN's 11-case regression suite is a standard three-dimensional validation reference, and every helium comparison against it inherits this 11% difference in α_A(He I). A disagreement in a helium ionization front or a helium line intensity against MOCASSIN may therefore be largely a choice of recombination data set rather than a transport or geometry difference. MoCHII sits with the clustered three.

Table 1: Photon-energy band for a blackbody source. "Set where" distinguishes a hardcoded band from one the user can change.

code	band	set where	FUV?
Wood `ionize`	13.6–54.4 eV	`planckset.f`, hardcoded	no
CMacIonize	13.6–54.4 eV	`PlanckPhotonSourceSpectrum.cpp`, hardcoded	no
MOCASSIN	1.4×10^{-4}–204 eV	`nuMin`, `nuMax` in the input	yes
TORUS	$\geq 1.2 \times 10^{-3}$ eV	`photoion_mod.F90`, hardcoded	yes
MoCHII	4.78–100 eV[a]	`par%efuv_min`, `par%eion_max`	default[a]

[a]Since 2026-09-25 `par%add_fuv` follows the He I 2^3S population when the input leaves it out: with helium and `par%hei_metastable_coupling` (the default) the band starts at `par%efuv_min` = 4.78 eV; otherwise at `par%eion_min` = 13.598 eV (6 eV with an explicit `add_fuv`). When this report was written the band was 13.598–100 eV and the extension optional.

9. TORUS

`spectrum_mod.f90:getWavelength` draws r, locates it in `spectrum%prob`, and linearly interpolates adjacent tabulated wavelengths; `photon_mod.f90:initPhoton` uses it. Searches of active `torus/src` found no Wood/CMacIonize-like H–He reemission handler, temperature-indexed free-bound LyC CDF, or HII recombination-ionizing-packet emission call. This is an absence of exposed HII diffuse-LyC physics, not evidence for a threshold approximation.

10. Where each code cuts the band

Every one of these codes fixes a lower and an upper photon energy and samples the blackbody only between them; none integrates over $(0, \infty)$. What differs, by orders of magnitude, is where the lower cut sits, and with it whether the non-ionizing continuum exists in the calculation at all.

Wood and CMacIonize: the ionizing band only. `planckset.f` sets `numin=3.289e15` Hz, the Lyman limit, and `numax=4` *in units of numin*, then lays the grid down as `nu(i)=1.+(i-1)*(numax-1)/(nfreq-1)`. The band is therefore 1 to 4 Ryd, 13.6 to 54.4 eV — it stops at the He II threshold and never reaches below the hydrogen one. CMacIonize copies it, comments included: `max_frequency = 4.` carries "in units 13.6 eV (corresponds to 54.4 eV)" and `min_frequency = 3.289e15` carries "13.6 eV in Hz".

MOCASSIN: the full continuum, and the stellar packets are launched across it. `set_input_mod.f90` defaults to `nuMin` = 1.001×10^{-5} and `nuMax` = 15 Ryd, both readable from the model input. A wide grid alone would not settle the question, since a code can tabulate a continuum for its dust and diffuse output while launching only ionizing packets; MOCASSIN does not. Its emission CDF is accumulated from the first bin, `inSpectrumProbDen(iS,i) = inSpectrumProbDen(iS,i-1) + inSpSumErg(i)*widFlx(i)/normConstantErg` over `i = 2, nbins`, and only the photon-count normalization `normConstantPhot` starts at the Lyman-limit bin `lymanP`. Non-ionizing photons are emitted and transported.

TORUS: the same, from the sub-millimeter up. `photoion_mod.F90` sets `nuStart = cSpeed/(1000.e4*1.d-8)`, a wavelength of 1 mm, and `nuEnd = 2*maxval(grid%ion%nuThresh)`, twice the highest threshold in the active ion list, with 1000 logarithmic bins between them. Packets are drawn over the whole range and the code counts which of

them are ionizing; the branch that would have skipped the rest, `if ((thisFreq*hcgs*ergtoev) < 13.6) cycle mainloop`, is present but commented out, so the choice to transport them is deliberate.

MoCHII: the band below the hydrogen edge follows the He I 2^3S *population.* When this report was written the default `par%eion_min` = 13.598 eV placed MoCHII with Wood and CMacIonize, and `par%add_fuv` (off) added `nnu_fuv` = 8 tally bins down to `par%efuv_min` = 6 eV. Since 2026-09-25 an input that leaves `add_fuv` and `efuv_min` out gets the extension from 4.78 eV whenever the 2^3S population is solved (`hei_metastable` or `hei_metastable_coupling`, the latter on by default, with `abund_He` > 0), and the FUV packets are launched as their own stratum (`par%fuv_sampling = 'separate'`, `no_photons`/4 by default). So a default MoCHII run is banded like MOCASSIN and TORUS from 4.78 eV, not like Wood.

What the lower cut excludes. A band floor at the hydrogen threshold is not a matter of resolution: it removes physics. Every excited level of He I has an ionization potential below 13.6 eV — 2^3S at 4.767, 2^1S at 3.971, 2^3P at 3.623 and 2^1P at 3.369 eV — so a code banded from 13.598 eV upward cannot photoionize any of them, whatever their populations. MoCHII says so at run time when the 2^3S population is solved and an input sets the floor above 4.78 eV (`setup.f90`), `NOTE: hei_metastable band floor 13.598 eV > 4.78 eV -> Phi_3 misses [4.78, floor); set add_fuv with efuv_min = 4.78 . . . .` Wood and CMacIonize share the limitation and do not report it; MOCASSIN and TORUS do not have it. The estimate of §11. put the missing 2^3S channel at 0.2–0.6% of the ground-state ionization rate for the Lexington HII40 conditions (made 2026-07-30 with photons above 13.598 eV only; the code's own note is that for a 40 kK star the unattenuated 4.78–13.6 eV part of the 2^3S photoionization integral is $\sim 10\times$ the part above 13.6 eV, so that estimate is low and has not been redone) — and the calculations where sub-Lyman photons do work (photodissociation regions, the metastable diagnostics) are exactly the ones a 13.6 eV floor silently truncates.

11. Implication

Dated (2026-07-30). Every He I front position in this section was measured on runs stopped by the earlier test on the $x_{\rm HII}$ step alone, which ended them while the fronts were still advancing. Run to the drift stopping test of 2026-09-25 (`par%conv_crit = 'vol'`, `fixed_point_distance_mod`), the HII40 He I front sits at 4.2719 pc against Cloudy's 4.305 pc (`results/continuous_energy/README.md`, `m8_defaults`), a residual of 0.033 pc: most of the 0.18 pc analyzed below was premature stopping. The sensitivities below are a record of those runs, not current bounds.

MoCHII already evaluates transport cross sections at each sampled source energy, unlike frequency-pointer approaches. Its ground-state diffuse continuum, formerly the distinct limitation, now follows the Milne relation with the same threshold-aware cross section transport uses, interpolating the CDF in temperature before a single inversion — the prescription §12. argues for, and one step beyond both Wood (nearest temperature row, grid frequency) and CMacIonize (interpolation between frequencies drawn from two temperature rows). The measured outcome confirms the prediction made here: the converged HII40 calculation moves the He I front by only +0.00063 pc, closing 0.3% of the Cloudy residual. The exact sampler was physically required and it did not remove the residual. Three further corrections have followed inside the He I excited cascade: the Drake two-photon shape, −0.00030 pc; one consistent set of branch coefficients, +0.00024 pc; and the 584 Å escape-probability split, −0.00363 pc, the largest of the four and six times the Milne step. Four independent improvements to what the diffuse field *emits* are therefore worth −0.00306 pc in total,

while the channel's mere presence is worth 0.10 pc: $r(\mathrm{He}^0 = 0.5)$ moves from 4.12333 to 4.11970 pc against the Cloudy c25.00 value of 4.30531 pc, so the sum is 1.7% of the 0.18 pc gap and points away from Cloudy rather than toward it. Each of the four is separately required — the acceptance criterion is physical correctness, not agreement with a particular reference calculation — and together they settle the diagnosis: **the residual is not in what the diffuse field emits.** The shape of the emitted spectrum, its branch weights and the fate of its strongest line have each been corrected independently, against an external implementation or an exact relation, and their sum is negligible and of the wrong sign. The two candidates named next in earlier versions of this report — the *transport* of those photons and the He I ground-continuum atomic data — have since been bounded quantitatively, and both are excluded.

Transport. The diffuse treatment can be switched between its two extremes, explicit transport of the packets and on-the-spot absorption at the emission site, and the front moves from 4.11970 to 4.18159 pc. The entire dynamic range of that choice is 0.062 pc, and Cloudy sits 0.124 pc beyond the more favorable end of it. Since the four reviewed codes differ precisely in where on this axis they sit — Wood and CMacIonize destroy their reemitted photons locally, MOCASSIN carries a diffuse population, MoCHII transports both the continua and the He I cascade products — the comparison of §7. spans the whole axis, and no position on it reaches the residual.

The α_1 cross-check, and a defect it exposed. MoCHII's He I cross section reproduces Verner's own `phfit2.f` — the same routine Wood's `probinit.f` calls to build its Milne CDF — to a ratio of 1.00000 at every energy over 24.6–105 eV, and its threshold value of 7.436 Mb sits 0.5% above the measured 7.40 ± 0.07 Mb of Samson et al. (1994), inside the experimental uncertainty. Because `recomb_mod` formed $\alpha_B = \alpha_A - \alpha_1$ with α_A from the Badnell total and α_1 from the level-resolved Mao & Kaastra fit, the "rates" side of the cross-check *is* α_1(He I), so its 3.44% separated two determinations of one number — and the subtraction crossed two unrelated sources.

Cloudy c25.00 settles which side is right, because it derives its H-like and He-like recombination level by level from its *own* photoionization cross sections through the exact Milne relation (`source/iso_radiative_recomb.cpp`), caching the result in `data/h_iso_recomb.dat` and `data/he_iso_recomb.dat`; row 0 of each element is the ground level and the last row the total, verified on hydrogen, where row 0 is 1.5840×10^{-13} and the last row 4.1642×10^{-13} against the known $\alpha_{1s} = 1.58 \times 10^{-13}$ and $\alpha_A = 4.18 \times 10^{-13}$ cm^3 s^{-1}. At 10^4 K, in 10^{-13} cm^3 s^{-1}, α_1 is:

T	BSS99 + JSM	Mao & Kaastra	Milne integral	Cloudy c25.00
5000 K	2.157	2.184 (+1.3%)	2.259 (+4.8%)	—
10^4 K	1.540	1.565 (+1.6%)	1.609 (+4.5%)	1.603 (+4.1%)
2×10^4 K	1.100	1.128 (+2.6%)	1.151 (+4.7%)	—

Earlier versions of this report read the first three columns as a majority vote and concluded that **the Milne integral was the outlier**. **That conclusion is withdrawn.** The arithmetic was right — the Milne integral does sit 4.5–4.8% above BSS99 and 2.8% above Mao while the two fits cluster within 2.6% — but the mutual closeness of two fits is not evidence that either reproduces the cross sections in use, and Cloudy, computing the same quantity level by level, lands within 0.34% of the Milne integral. Across the three channels at 10^4 K, Badnell's totals are accurate to 0.3–0.7% and the Mao $n = 1$ values to 0.2–0.3% for H I and He II, but He I is 2.38% low, which the subtraction propagated into α_B(He I) as +1.88%. MoCHII's own Milne integral matches Cloudy to 0.3%, 0.34% and 0.04%.

The α_1 coefficients were accordingly regenerated from the Milne relation evaluated on the cross sections the transport

actually uses, keeping the same seven-coefficient functional form so that no code structure changes (fit residual $\leq$ 0.019% over 10^3–10^5 K). All three channels now agree with Cloudy to $\leq 0.35\%$, He I moving from −2.38% to +0.34%, and the start-up cross-check that reported 3.44% reports 0.01%. The atomic-data inconsistency this report previously described as inherited by every code that pairs Verner cross sections with recombination fits was in fact a source-mixing defect in one of them. Its measured effect is small and in the helpful direction: $r(\mathrm{He}^0 = 0.5)$ moves +0.00811 pc at HII40 and +0.00210 pc at HII20, and the Cloudy residual becomes −0.17750 pc. $L(\mathrm{H}\beta)$ falls 0.37–0.42%, i.e. slightly further below the published values — recorded rather than buried, and not an argument against a change whose criterion is an exact relation.

Metal opacity. The one candidate outside the helium data has now been measured as well, and it is also excluded. Scaling the summed metal photoionization cross section on the converged HII40 model moves the He I front by +0.03093 pc at $\times 1.30$ and −0.07206 pc at $\times 0.70$, and switching metal absorption off entirely costs 0.615 pc. The response is strongly asymmetric and saturating — 0.615 pc per unit downward against 0.103 pc per unit upward — so closing 0.178 pc would take +180% metal opacity, and more once the saturation is allowed for. The mechanism is spectral hardening rather than opacity magnitude: stages with thresholds in 21–40 eV (Ne I 21.56, S II 23.33, C II 24.38, N II 29.60, S III 34.83, O II 35.12 eV) remove the softest He-ionizing photons, and $\sigma(\mathrm{He\,I})/\sigma(\mathrm{H\,I})$ rises from 6.1 at 25 eV to 19.9 at 95 eV, so the surviving field is taken up by He^0 rather than H^0. MoCHII's metal cross sections reproduce Verner's `phfit2.f` to 10^{-6}, and an Opacity Project comparison runs 2–14% *lower* for C II, N II and O II, i.e. the wrong direction. An earlier estimate here that a $\sim 30\%$ metal-opacity error would be worth ~ 0.18 pc extrapolated the on/off bound as if the response were symmetric; it is withdrawn — 30% is worth 0.031–0.072 pc.

What is left. With dust off by construction and the source photon budget matching Cloudy to 0.003% in both $Q(>$ 13.6 eV) and $Q(>24.6$ eV), the residual is unexplained rather than understood. Its signature is helium-specific and cumulative: $x(\mathrm{H}^0)$ agrees with Cloudy to 1–2% throughout, while the $x(\mathrm{He}^0)$ ratio grows monotonically outward from 1.007 at 1.6 pc to 1.196 at 3.8 pc, which is what a small seed must do when He^0 carries 84–96% of the opacity above 24.6 eV and 24.4% of the He-ionizing supply is lost to competing absorbers. Closing 4.1% in radius means 13% in volume, i.e. 14% in $Q(\mathrm{He}^0)$ or 20% in α_B(He I), and neither is available: scaling σ_{pi}(He I) by −3% moves the front only +0.00799 pc and the He I recombination rate by −3% only +0.02410 pc, so 0.178 pc would take 70% and 23%, and MoCHII's α_A, α_1 and α_B(He I) now sit within 0.3%, 0.35% and 0.3% of Cloudy's own level-resolved values (α_B was 1.9% off before the α_1 correction above; current values from `tools/python/mochii_rates.py`, which reads the same coefficients as `recomb_mod`). The helium *network* candidate named in earlier versions of this report has been examined in §8. and does not supply it either: no code truncates its cascade, MoCHII agrees with Cloudy's explicit level sum to 0.3%, and the He I excited levels that a resolved model atom carries lie below MoCHII's 13.598 eV band floor and are worth 0.2–0.6% of extra ionization. No single parameter remains. What has not been tested is a combination of several few-percent differences — in the He I recombination rate, in the shape of σ_{pi}(He I), in the metal opacity of the 21–40 eV window, in Cloudy-side modeling choices not yet identified — acting together through the He^0 optical depth, which compounds them with depth exactly as the radial signature shows. That needs a joint experiment, not a one-at-a-time one.

12. Which method is physically most accurate?

The most accurate practical Monte-Carlo prescription is *not* a fixed energy, an exponential tail, or merely a very fine arbitrary energy grid. For each recombining ion, electron temperature, and (where required) density, construct the photon-number PDF from the level-resolved free–bound emissivity obtained by detailed balance,

$$p_i(\nu \mid T_e) \propto \frac{j^{\mathrm{fb}}_{\nu,i}(T_e)}{h\nu}, \qquad \nu \geq \nu_{0,i},$$

using the same resolved photoionization cross section in the Milne relation that is used to absorb the packet. The emissivity should include the relevant free–bound Gaunt factor and should be integrated with accurate quadrature; its CDF and its mean photon energy must be derived from that same PDF. Branching to recombination levels and subsequent He I/He II cascades must be selected from level-resolved effective coefficients, with the full two-photon spectral shape rather than a flat surrogate. This prescription is exact subject to the adopted atomic data and the assumption of a Maxwellian electron distribution. Wood ionize and CMacIonize are closest among the reviewed codes: both use $\nu^3\sigma(\nu)\exp[-h(\nu-\nu_0)/kT]$ and temperature tables. CMacIonize improves the source CDF interpolation, but its LyC implementation interpolates frequencies selected from two neighboring temperature CDFs rather than directly in a temperature-interpolated CDF; Wood uses the nearest temperature row. Therefore neither is the final reference implementation. MoCHII now does interpolate the *CDF* in temperature, invert once, sample a continuous energy, and carry that same energy into the existing exact transport cross sections. Its He I cascade now uses one consistent set of effective recombination coefficients, the measured two-photon shape and the same 584 Å escape-probability split as Wood and CMacIonize, applied before transport rather than in place of it (§7.), so the remaining approximations there are explicit resonance-line transfer of the trapped 584 Å photon and collisional redistribution among the $n = 2$ terms, not the shape of the emitted continua.

13. Why can Wood ionize be fast?

The accuracy of the diffuse-LyC *spectral shape* is not the dominant cost. Wood ionize precomputes its small T–ν Milne CDF arrays in `probinit.f`; during transport, `nuemit.f` only draws a uniform deviate and performs a table search. It does not numerically integrate the Milne relation or evaluate a new continuum spectrum for every emitted packet.

More importantly, Wood uses event-by-event reemission. Following absorption, `mcionize.f` selects absorption species and reemission channel; a diffuse packet is created only when that event branches into an ionizing reemission. MoCHII instead rebuilds an explicit volume-emissivity CDF every iteration and transports a separate diffuse population whose size is $L_{\mathrm{diffuse}}/L_{\mathrm{packet}}$. In HII40 this was commonly 10^7–2×10^7 diffuse packets per iteration on the production inputs of 2026-07-30, in addition to direct packets.

Wood also confines this benchmark physics largely to H and He and samples a modest frequency grid. MoCHII additionally pays for AMR leaf traversal, continuous-energy H/He/metal path scoring, metal ionization cascades and cooling, thermal equilibrium iterations, and HDF5 diagnostics. These choices can be scientifically necessary, but they dominate runtime relative to one inverse-CDF operation.

Consequently, adding a precomputed Milne CDF to MoCHII should have negligible direct sampling cost. It should not be expected by itself to make MoCHII as fast as Wood: the major cost is explicit diffuse-packet transport and the

broader MoCHII physics. The correct comparison is therefore limited: Wood is closest for the H/He ground-continuum spectral PDF, but its nearest-temperature, grid-frequency sampling is still less rigorous than a MoCHII CDF/PDF-temperature interpolation followed by continuous-energy inversion.